

Passphrase Lab



BIG IDEA Length beats cleverness. A long, random passphrase is much harder to guess than a short, tricky password.

1 HOOK 2'	2 ACTIVITY 8'	3 DISCUSS 3'	4 TAKEAWAY 2'
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PREP IN 5 MINUTES

Materials: Passphrase Word List (R-2.1, 36 words), two dice per group, Crack Timer Worksheet (p. 4). No real accounts or passwords.

- One Word List and two dice per group.
- Copy the Crack-Timer table (R-2.2) on the board; cover the last column.
- Draw a robot: the Guess-Machine.

1 HOOK 2 MIN *60-second script* “Mateo from our squad has a secret clubhouse. He locked it with a number lock with just ONE number, zero through nine. Priya said, ‘I bet I can open it in ten tries.’ And she did! So Mateo made a lock with TWO numbers: up to a hundred tries. Then THREE numbers: up to a thousand tries. Every extra number makes the lock harder. A computer can try thousands of guesses a second, so the squad needs a lock that is long as well as clever. Today we run a Passphrase Lab. Lab coats on, hackers. Well, imaginary ones.”

2 ACTIVITY 8 MIN *The Crack-Timer Lab*

- (1 min) Meet the Guess-Machine.** It never tires and makes 1,000 guesses a second in our lab (a gentle model).
- (2 min) Reveal the Crack-Timer table** row by row while the class says the multiplication aloud. Each new word multiplies the lock by 36. “Length wins.”
- (3 min) Roll a practice passphrase.** First die is the row, second die the column; roll three times. “Random is stronger than clever.” Practice only, never for a real account.
- (2 min) Do the math.** 3 words from 36 is 46,656 possibilities, about 47 seconds. Real lists have 7,776 words, so four words give over 3 quadrillion.

3 DISCUSS 3 MIN

- “Why did we use dice instead of picking our favorite words?”
- “Who should know your real passphrase?” (Only you and your parent or guardian.)
- “What if a website or a person asks for your password?” (Pause. Check. Tell.)

4 TAKEAWAY 2 MIN “If someone asks for your password, your hackles should go up. A false alarm is fine. Long and random beats short and clever. Share your passphrase only with a parent or guardian.”

Hand signal: Hackles Up (hand brushed up the neck), Pause (palm), Check (magnifier fingers), Tell (hand on chest, then out).

EXIT TICKET “Which is harder to guess: a 3-word dice passphrase or ‘Dragon1’? Circle one and tell why in one sentence.”

FUN BOOST Human Guess-Machine (30 seconds). A volunteer robot-walks (“BEEP BOOP”) and guesses a classmate’s secret number, one digit and then two, while the class counts the guesses.

LEVEL-UP Beat the Machine. How long for an 8-word passphrase from the 36-word list? (About 89 years. Just a classroom model.)

DIFFERENTIATE Pre-fill the Crack-Timer table; pair students for rolling; add pictures to the word list.

FAMILY CARD Roll dice to build a family passphrase for a shared account. Keep it private and turn on two-step verification.

IF A STUDENT TELLS YOU SOMETHING

• Say: “Thank you for telling me. You did the right thing.” • “This is not your fault, and you are not in trouble.” • “I can’t keep this a secret, because I need to get you help. I will tell you who I tell.”	• Do not say: “I promise I won’t tell anyone.” • “Are you sure?” or “Why would you give it away?” • Anything beyond “Can you tell me more?”	First 10 Minutes: see the Disclosure Guide. Then: tell your counselor or designated person right away. Follow your school policy and your state’s reporting law.
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